Differentiation

The chain rule

1 Find $\dfrac{dy}{dx}$ for $y = (3x+1)^5$.

2 Find $\dfrac{dy}{dx}$ for $y = \sqrt{4x^2+1}$.

3 Find $\dfrac{dy}{dx}$ for $y = (2x-5)^6$.

The product rule

4 Find $\dfrac{dy}{dx}$ for $y = x^2 \sin x$.

5 Find $\dfrac{dy}{dx}$ for $y = e^{3x} \cos x$.

The quotient rule

6 Find $\dfrac{dy}{dx}$ for $y = \dfrac{x^2}{2x+1}$.

7 Find $\dfrac{dy}{dx}$ for $y = \dfrac{\sin x}{x}$.

8 Find $\dfrac{dy}{dx}$ for $y = \dfrac{e^x}{x^2+1}$.

Implicit differentiation

9 Find $\dfrac{dy}{dx}$ given $x^2 + y^2 = 25$.

10 Find $\dfrac{dy}{dx}$ given $x^3 + y^3 = 6xy$, and evaluate it at the point $(3, 3)$.

11 Find $\dfrac{dy}{dx}$ given $y^2 - xy + x^2 = 7$, and evaluate it at the point $(1, 3)$.

12 Find $\dfrac{dy}{dx}$ given $\sin(xy) = x$, using implicit differentiation.

Parametric differentiation

13 A curve is defined parametrically by $x = t^2$, $y = 2t^3$. Find $\dfrac{dy}{dx}$ in terms of t.

14 A curve is defined by $x = 3\cos t$, $y = 3\sin t$. Find $\dfrac{dy}{dx}$ in terms of t, and find the gradient at $t = \dfrac{\pi}{4}$.

Related rates

15 A spherical balloon is inflated so its volume increases at a constant rate of 100 cm^3/s. Find the rate of increase of the radius when $r = 5$ cm, using $V = \frac{4}{3}\pi r^3$, to three decimal places.

16 The area of a circular ripple increases at 8π cm^2/s. Find the rate of change of the radius when $r = 4$ cm, using $A = \pi r^2$.

17 A cone-shaped tank (vertex down) has height 10 cm and top radius 4 cm, so at water depth h the radius is $r = \frac{2h}{5}$. Water is poured in at 12 cm^3/s. Find the rate of rise of the water level when $h = 5$ cm, using $V = \frac{1}{3}\pi r^2 h$.

18 The radius of a circular oil slick increases at a constant rate of 0.5 m/min. Find the rate of increase of its area when $r = 20$ m.

Differentiating trigonometric, exponential and logarithmic composites

19 Find $\dfrac{dy}{dx}$ for $y = \sin(3x^2)$.

20 Find $\dfrac{dy}{dx}$ for $y = \ln(5x^2 + 1)$.

21 Find $\dfrac{dy}{dx}$ for $y = e^{\sin x}$.

22 Find $\dfrac{dy}{dx}$ for $y = \ln(\sin x)$.

Applying the rules: tangents and second derivatives

23 Find the equation of the tangent to $y = xe^{-x}$ at $x = 1$.

24 Find $\dfrac{d^2y}{dx^2}$ for $y = \ln(x^2 + 1)$, and evaluate it at $x = 1$.

A well-designed word problem: a ladder sliding down a wall

25 This question has three parts. A ladder 5 m long leans against a vertical wall, its base on horizontal ground. The base slides away from the wall at a constant rate of 0.4 m/s. (a) Let x be the distance of the base from the wall and y the height of the top of the ladder. Write the relationship between x and y, and differentiate implicitly with respect to time t to relate $\frac{dx}{dt}$ and $\frac{dy}{dt}$. (b) Find the rate at which the top of the ladder is sliding down the wall when the base is 3 m from the wall. (c) Using $x = 5\cos\theta$, where θ is the angle between the ladder and the ground, find the rate at which θ is changing at that instant.